

Trust Enhancement Scheme for Cross Domain Authentication of PKI system

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Abstract—Public Key Infrastructure (PKI) has been popularized in many scenarios such as e-government applications, enterprises, etc. Due to the construction of PKI system of various regions and departments, there formed a lot of isolated PKI management domains, cross-domain authentication has become a problem that cannot be ignored, which also has some traditional solutions such as cross-authentication, trust list, etc. However, some issues still exist, which hinder the popularity of unified trust services. For example, lack of unified cross domain standard, the update period of Certificate Revocation List (CRL) is too long, which affects the security of cross-domain authentication.

In this paper, we proposed a trust transferring model by using blockchain consensus instead of traditional trusted third party for e-government applications. We exploit how to solve the unified trust service problem of PKI at the national level through consensus and transfer some CA management functions to the blockchain. And we prove the scheme's feasibility from engineering perspective. Besides, the scheme has enough scalability to satisfy trust transfer requirements of multiple PKI systems. Meanwhile, the security and efficiency are also guaranteed compared with traditional solutions.

Keywords—Blockchain; PKI (Public Key Infrastructure); Trust delivery; E-government; Cross-domain; Authentication.

I. INTRODUCTION

PKI (Public Key Infrastructure) technology plays an important role in information security as an infrastructure that can realize core security services such as identity authentication, confidentiality, integrity and non-repudiation. With continuous maturity of technology, a large number of CA (Certificate Authorities) centers are constructed in each country, and almost all of them are faced with interoperability issues. The lack of top level design has led to blindness in development and many problems, the most serious of which is the interface problem. Different CA trust domains cannot authenticate each other, which also leads to the problem of limited subsequent development.

Currently, there are mainly three methods to solve cross-domain problems for PKI systems. Cross authentication between CAs, Trust list, and Bridge CA [1]–[4]. Cross-authentication enables cross-domain authentication of users in different domains by issuing cross-certificates between CAs. That is, one CA uses its own private key to issue digital certificates to the other CA to determine whether its

domain trusts the other, vice versa. The trust list is a data structure that contains a list of "trusted CAs" after signing by trusted third party. When a domain user authenticates another domain user, it checks whether the anchor CA of the other party is in the trusted list to determine whether the identity of the other party can be trusted. Bridge CA (BCA) is not a traditional tree-like CA, nor a mesh CA. It doesn't directly issue certificates to users. It isn't a unique trust point like a root CA. It is just a single CA and establishes a two-way trust relationship with different CA trust domains.

Cross-certification between trust domains is easy to form a complex mesh topology, which is difficult to use widely in e-government. The most common usage of trust list is Web Browser [5]–[7], the browser vendor usually install several anchor CAs in browser, however, it is difficult to maintain the trust list dynamically after release if one of them is "bad" or need to add some extra anchor CAs to the list. U.K government adopted scheme Verify-Hub to provide identity services for public, Verify-Hub is application-centric and combines multiple designated identity providers (a kind of trust list maintained by government). User can choose different authentication service providers for authentication and login, the authentication result is then passed to government service application by providers. In this way, it realized unified authentication and protected user privacy, but single sign-on and off-site authentication are not considered. Considering that China has far more populations and information systems than UK, national and provincial information centers do not have strict affiliation. It is almost impossible to adopt the British type because of huge cost and lack of affiliation. Currently, the successful bridge CA only has the U.S Federal bridge CA [8], [9]. The U.S Federal Bridge CA has broken many CA domains including complex tree, mesh and trust list structures, including not only government agencies at all levels, but also many business partners. Its success is the embodiment of the national will, and it is difficult for private commercial organizations to apply. The Canada and Korea's e-government PKI systems are the typical central tree structure [10], its solution is suitable for sparsely populated, small countries, thus has less reference to China's national CA construction management. Japan's GPKI (Government PKI) [11] is mainly composed

of CA maintained by Ministry of Internal Affairs and Communications and CA maintained by Ministry of Justice, which also includes Bridge CA.

China's PKI system is similar to Japan's [12], [13] in some respects. The division of regions and industries is more detailed. Take a look at the current situation of PKI in China, the PKI system at the national level in China has not been fully established and PKI e-government has not been unified nationwide. Besides, the development between regions and industries is not balanced, the different CAs are independent of each other, there are even some conflicts, resulting in repeated certificate applications and high idle rate. Some districts have already realized comprehensive networking while some others have not. It is urgent to realize trust transfer between different PKIs in China.

Meanwhile, there are also some blockchain-based PKI solutions. For example: CertCoin [14] canceled the CA Certificate Authority, using the blockchain as a distributed ledger for domain names and public keys, providing auditing functions and PKI solutions without single point of failure; Ukrainian REMME [15] proposed a scheme that gives each blockchain device a unique SSL certificate to eliminate the possibility of forging a certificate; Estonia Guardtime [16] uses the blockchain to create a KSI Keyless Signature Infrastructure instead of PKI, which has already recorded over 1 million people in Estonia and becomes a mature blockchain company. All these schemes focus on the construction of decentralized PKI, and even completely change the authentication principle of traditional PKI without improving the existing centralized PKI.

In this paper, we propose a blockchain-based trust transferring scheme for different PKI systems to realize trust enhancement and transfer. Our contributions are summarized as follows:

- We propose a trust transferring model by using consensus of blockchain instead of traditional trusted third party for e-government applications.
- we exploit how to solve the unified trust service problem of PKI at the national level through consensus and transfer some CA management functions to the blockchain.
- we proved the feasibility of our proposed scheme theoretically. Besides, based on consortium blockchain, our scheme has enough scalability to satisfy trust transfer requirements of multiple PKI systems. Further experiments show that our scheme has the characteristics of high efficiency and supervision in terms of unified authentication compared with traditional schemes.

The rest of the paper is organized as follows:

The remainder of this paper is organized as follows. In Section II, we introduce the preliminaries, which includes cross-domain theory, Certificate Revocation List (CRL) and Online Certificate Status Protocol. After that we outline the reform of trust mechanism in our scheme. In Section III,

we describe the system model, cross domain authentication structure and trust delivery requirements. Then we design the cross-domain protocol in detail in Section IV and carry out the security analysis in Section V, respectively. Finally, Section VI concludes the paper.

II. PRELIMINARIES AND REFORM OF TRUST MECHANISM

A. PKI cross-domain authentication theory

As is shown in Fig. 1. Cross-authentication is a CA (assumed to be CA1) to sign the root certificate of another CA (assumed to be CA2). It can implement cross-domain authentication for users in CA2 domain to access the CA1 domain. CA1 is required to securely transmit the public key used to verify the signature to CA2. Use CA1's own private key to issue a digital certificate to the other party (CA2) determines whether the domain (CA1) trusts the other party's domain (CA2). In practical applications, when a user (authenticated party) needs cross-domain access, it is necessary to present a certificate to the relying party to verify its identity. When the relying party cannot verify the validity of the authenticated party's identity, some additional certificates are required to verify the other party's credibility. A series of certificate sets from the CA certificate trusted by the relying party to the user certificate to be verified is called the certificate chain or certificate path, and the verification process is called cross-authentication.

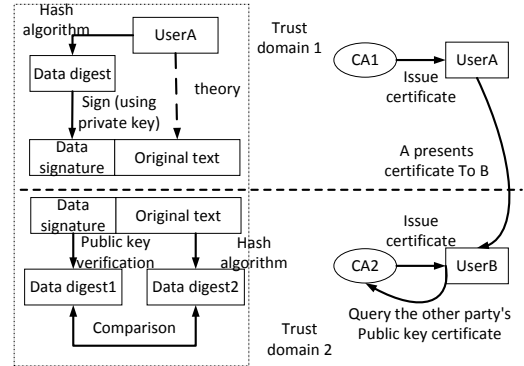


Figure 1. The cross-domain authentication theory in PKI

PKI cross-domain authentication theory is shown in fig. 1. When user A belonging to CA domain1 wants to communicate with user B belonging to CA domain2. User A will first hashes the plaintext as a digest, he then signs the digest with the private key and sends it to user B along with the plaintext. User B receives the signature and the plaintext, he will first searches for the cross certificate containing the A public key at the root CA of domain2, verifies the signature with user A's public key, The plaintext is processed with the same hash algorithm and then compared with the digest verified with the public key. If it is the same, the trust relation with A will be achieved.

B. CRL (Certificate Revocation List)

CRL in traditional PKI is used to retrieve whether the certificate is revoked or not, prompting the transaction party not to use the public key bound by the certificate in the CRL [17], [18]. Equivalent to the certificate blacklist, once the party is in the list, it cannot be verified (the CRL data structure of the X.509 standard is shown in Table. I).

The extensions include: the code for revocation reason, certificate issuer, instruction code for temporary freezing, expiration date, etc. The use of traditional CRL faces the problem of long list release time, large storage overhead and large transaction concurrency resulting in reduced efficiency. To solve the above problems, the traditional solutions are as follows:

- 1) Divide the large CRL into multiple controllable fragments and publish the certificate cancellation information of one CA through multiple CRLs. In the extension of the certificate, set the CRL distribution point sub-item, mark the location of the CRL distribution point, and guide the user to access the corresponding CRL according to the parameters.
- 2) Reduce the concurrent access pressure by setting the LDAP (Lightweight Directory Access Protocol) mirror server.

C. OCSP (Online Certificate Status Protocol)

In order to make up for the shortcomings of CRL's real time performance, large network bandwidth resource consumption and low verification efficiency, the traditional scheme uses OCSP (rfc2560 defines the message format for OCSP client and servers) [19]–[21] to query the status of the certificate in real time.

D. Reform: BCRL (Blockchain-based Certificate Revocation List) with OCSP

In our scheme, the blockchain account can play the role of the certificate blacklist. Regarding the real-time status of the certificate, since the blockchain ledger only stores the data's hash value, it only provides the authenticity of the message in terms of query, and cannot return specific plaintext information. Therefore, OCSP is still used as a supplement to the BCRL in our scheme. The CA issues the certificate status to the LDAP server, The engineering implementation can be queried by the authentication server or the smart contract. In our scheme, the BCRL-based query strategy and the BOCSP-based accountability mechanism can distribute the bandwidth resource overhead to each independent authentication system and blockchain node, which will greatly improve the certificate verification efficiency.

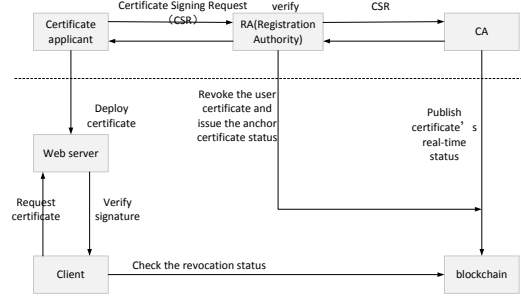


Figure 2. Blockchain-based certificate Life Cycle

III. SYSTEM MODEL, CROSS-DOMAIN AUTHENTICATION STRUCTURE AND TRUST DELIVERY REQUIREMENTS

A. System Model

Our trust transfer scheme is constructed on consortium blockchain, targeting only members of a specific group and a limited number of third parties (such as regulatory parties). Bookkeeper nodes can be designated and they jointly maintain the ledger and block generation.

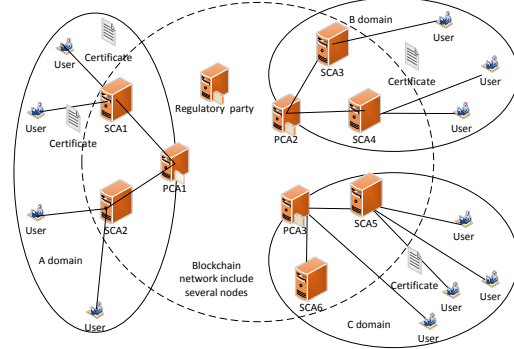


Figure 3. System model

As is shown in fig. 3, two levels of authentication domain are considered in our model.

- **Trust domain:** Each trust domain takes its PCA (Prime Certificate Authority) as its trust anchor. PCA, a number of SCAs (Subordinate Certificate Authority) and issued users together form a trust domain (circle of solid line in fig. 3, domain A, B and C).
- **Trust federation:** the CA organization (all the PCAs and some SCAs) together with the supervisor (all with blockchain nodes deployed) form a trust federation (circle of dashed line in fig. 3), which is constructed on the consensus of blockchain. It has good scalability and can contain multiple trust domains.

Table I
DATA STRUCTURE OF X.509's CRL

CRL edition	CRL signature algorithm	CRL issuer	update time	next time update time	revoked cert list	extension field
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B. Cross-domain Authentication Structure

To better understand the system model, fig. 4 explain the authentication structure in detail. The newly designed authentication structure looks a bit like traditional trust list. The difference is that the original trusted third party is replaced by the consensus among the participants. All the CAs of PKI systems that join the federation are equipped with blockchain nodes, and the subordinate CAs (such as CA4 and CA5) can be selectly deployed according to the environment. For example, when users in subordinate CA domain have more cross-domain authentication requirements, the placement of the blockchain node will effectively decentralize the system query pressure and improve the certificate verification efficiency.

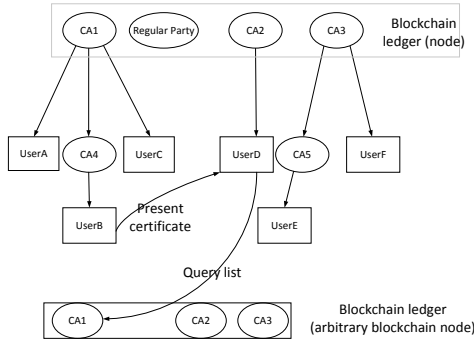


Figure 4. Cross-domain Authentication Structure

The consensus-based blockchain ledger also contains respective "trust list" issued by each PKI organization, called BCRL (Blockchain-based Certificate List), which contains the "certificate number" revoked by all federation members. Since the blockchain ledger itself only stores the hash value and doesn't store plaintext, BCRL can save more space than traditional CRL. Because the ledger won't return specific plaintext but valid or invalid information, it is necessary for the initiator to construct a corresponding revoked certificate data structure to obtain a return value for whether the constructed information is valid.

When user B presents his certificate to User D, it will be accompanied by a series of superior or cross certificates (the premise that the authentication can be pass is that CA1 and CA2 have already reached trust through negotiation, as is shown in point II-A). The validity of the certificate chain can be queried through any blockchain node. Since the BCRL release has a shorter cycle and the size is smaller, it is better than traditional CRL's hour-level period and

excessive list space. The accuracy of certificate verification will be effectively improved, at the same time, the real-time plaintext status of the certificates will be compensated using the original OCSP protocol.

C. Trust Delivery Requirements

Blockchain nodes are deployed at anchor CA agencies of all PKI participating in the federation, and the subordinate CA (SCA) agencies can choose to deploy according to the requirements (for example, if the SCA domains have more cross-domain requirements, the ledger can be deployed to disperse authentication pressure).

Federation-level deployment doesn't affect the authentication within original domain of each PKI trust domain. When cross-domain requirements are involved, trust delivery will be realized through the deployed trust interworking equipments (including blockchain, smart contracts, internal and external interfaces, etc).

Considering that the user exceeds the security boundary of the original authentication system when e-government application involves cross-domain authentication, there will be regulatory requirements at the unified authentication level. Therefore, a blockchain node is deployed to the supervisor and regulatory interface is set aside in the federation to present the cross-domain situation for the supervisor at the unified authentication level of the federation, so as to effectively monitor user behavior and trace responsibilities.

IV. CERTIFICATE AND PROTOCOL DESIGN

A. Certificate Description and Symbol Meaning

Currently, e-government in China mainly uses the X.509 standard certificate. Fig. 5 shows the contents of the X.509 V3 edition. Each certificate mainly consists of three parts: certificate attributions, signature algorithm and digital signature. The certificate attributions are further subdivided into required fields such as version number, serial number, algorithm ID, issuer name and optional fields such as issuer ID, Certificate ID and extension field. Required fields need to be identified in the certificate system authentication using the X.509 mechanism. The optional fields usually have some flexibility, they can be skipped if they are not recognized.

In order to be operational, the only difference is that when the original authentication system user needs cross-domain authentication, the intra-domain authentication server only needs to mark the cross-domain tag at the extension field of the certificate. The symbols' meaning are shown in Table. II

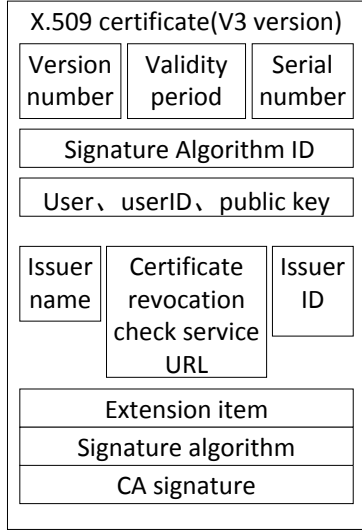


Figure 5. X.509 Certificate Description

Table II
SYMBOL MEANINGS OF THE PROTOCOL

Symbol	Meaning
nonce	Represents the session ID, associated with the client session, which mitigates replay attacks.
ReqCross	User initiates an authentication request to an out-of-domain application.
ReqStatus	The type of message request, such as the target domain when cross-domain.
$X.509_{federate}$	The extension field of X.509 is identified by the intra-domain authentication server, and other fields are the same as the original X.509.
$Cert_{chain}$	The certificate chain, including the user certificate and the certificate issued at each level, can be downloaded from the link in the user certificate attribute field.
DN	Distinguish Name, certificate unique alias.
$\{0,1\}$	Query message returned valid or invalid.
$SIG_X(*)$	Indicates that the participant X(if the X is not identified, the sender) signs all the previous data. If there is specific content in the parentheses, it means the signature of all the data in the brackets
Site	Message query destination address(Such as OCSP service address).
$\{DN/Cert\}_{chain}$	Array, list each certificate DN or certificate in the $Cert_{chain}$.
$\{0,1\}_{BCRL/Chain}$	Array, including whether each certificate is revoked(BCRLCert), or whether the $Cert_{chain}$ is valid(returned by OCSP) .
Auth-result	$Auth - result \in \{0,1\}$, indicates that the authentication succeeded or failed.

B. Protocol design

This protocol applies to the systems that use same certificate mechanism (X.509). User can access services of other PKI systems participating in the federation from the system to which they belong. When initiating an access, the user needs to present her certificate to the corresponding server.

The authentication server initiates a request to the blockchain according to the link of the presented certificate to verify whether each certificate in the chain is revoked, if not revoked, it will then verify signatures and others. The scheme uses the traditional OCSP protocol as a security supplement. In this protocol, the smart contract initiates a request to the OCSP server. The OCSP returns the real-time status of each certificate to the blockchain, as a supplement to the security strategies of certificate verification. The blockchain returns the information and time of each certificate of the certificate chain to the server. The authentication server confirms the information and passes the verification.

It is worth noting that, in order to ensure the efficiency of certificate verification, after the BCRL verification is verified, the blockchain returns the relying party of information about whether the certificate is invalid, and does not have to wait for the OCSP message, thereby improving the certificate verification efficiency. OCSP is only used as a security strategy for certificate validity, exist for auditing and retroactive responsibilities.

The protocol is as shown in Fig. 6

- 1) $user \rightarrow AS1: Nonce || reqCross || X.509 || SIG_{user}(*)$. The user submits a cross-domain application to the authentication server and presents his X.509 certificate.
- 2) $AS1 \rightarrow AS2: Nonce || reqCross || X.509_{federate} || SIG_{user}(*) || Cert_{chain} || SIG_{AS1}(*)$. The intra-domain authentication server AS1 verifies the user's cross-domain request, marks the X.509 extension field with the cross-domain federation identifier, construct the certificate chain, and forwards the message to AS2 (extra-domain authentication server).
- 3) Verify and check. AS2 verifies the signature of the message from AS1, including plaintext field of each public key certificate, such as: validity period, serial number, and so on.
- 4) $AS2 \rightarrow blockchain: Site || \{DN\}_{user} || \{Cert\}_{chain}$. AS2 sends the certificate chain and certificate DN (Distinguish Name) request to blockchain, querying the validity of the certificate chain (by querying OCSP server for each certificate's real-time status) and whether any of the certificate in the chain is revoked (by querying BCRL).
- 5) $blockchain \rightarrow OCSPserver : Site || ReqStatus || \{DN\}_{chain}$. The smart contract queries the OCSP server to check whether the certificate issued at all levels in the certificate chain is valid (this step can also directly initiate the query through the original authentication server).
- 6) $OCSPserver \rightarrow blockchain : DN || status_{chain} || SIG_{CA}(*)$. The OCSP server returns each certificate identifier and its status information to the blockchain, including "valid", "logout", "unknown", etc. and the corresponding verification time

- 7) 1) verify validity and signature of $Cert_{chain}$. 2) check validity of DN_{user} . The blockchain node receives the certificate status feedback, verifies whether the feedback message signature and the certificate status are available, if they are available, then queries whether any of the certificate DN number in the chain is revoked in the ledger.
- 8) $blockchain \rightarrow AS2$: $\{0,1\}_{BCRLCert} \parallel \{0,1\}_{Certchain}$. If the above certificate status is passed, the blockchain smart contract will return to AS2 whether the certificate chain is valid or not, as well as the status information and revocation information of each certificate.
- 9) Verify Certchain. The AS2 authentication server obtains the returned result and confirmed that the certificate chain status is valid. The following operations will be performed: verify the certificate signatures one by one according to the constructed certificate chain, the name, policy constraints, certificate key usage, extended key usage, etc.
Signature verification includes: (1) Unsealing certificate. Verify that the issuer CA's public key correctly unlocks the "issuer's digital signature" in the customer entity certificate. (2) Certificate chain verification, the process of certificate chain authentication and unsealing, starting from the root certificate (self-signed certificate) that can be identified by the verifier.
- 10) $AS2 \rightarrow user$ of $AS1$: (1) Auth-result. (2) $token \parallel timestamp \parallel SIG(*)$. After the certificate chain authentication is passed, AS2 returns a user authentication success message and issues an access token.

V. EFFICIENCY ANALYSIS

In the cross-domain transformation of the certificate system, the real-time validity of certificate validation is particularly important for information security. Traditional BCRL has the characteristics of excessive storage cost and slow update speed (usually several days), which may result in the revoked certificate still accessing the target system during the interval period. Some solutions use incremental CRL to solve the problem of wasted storage overhead, but the update cycle is still several hours long. This interval has a large security risk, which is enough for the malicious users who have been revoked to have sufficient time to successfully pass the target system authentication within the time interval. Incremental CRL reduces the storage cost of the list, thus obtaining a shorter time period. However, it takes only temporary measures but not permanent ones, and cannot update the CRL list immediately when every revocation is issued.

In contrast, verifying the certificate revocation information based on blockchain is far more efficient. The condition that cert revocation can be verified in the blockchain is that the

revoke information exists in the issued block. According to the fabric, a block will be generated when one of two conditions is met. One is that the number of transactions in the block reaches the upper limit of block allocation. Second, the number of transactions collected is insufficient, but blocks will also be generated when the time reaches the preset period. Both can be set according to transaction frequency and bussiness volume, and customized according to the actual needs. Based on the above characteristics of the fabric, when docking the certificate system, the release cycle of the revocation list (BCRL) based on blockchain is much shorter than that of the traditional CRL and incremental CRL. The update cycle reaches the order of 10 seconds, which effectively improves the validity of the cert verification when cross domain and authentication security. In practical application, the above indexes can be dynamically set according to the number of systems joining the alliance and transaction volume, thus saving resources and improving the utilization rate of the system.

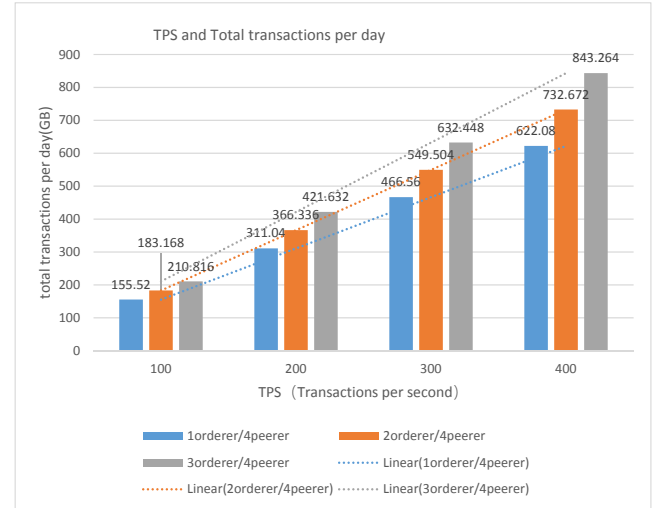


Figure 7. Comparison between TPS and total transactions per day

Considering that the issue efficiency of the blockchain is significantly different from that of the traditional scheme (on the order of 10 seconds and 10hours), this chapter only lists the storage overhead of the the blockchain certificate.

We have also verified the feasibility of the scheme. Experiment environment is shown in table.III. Figure. 7 shows the storage overhead changes of the system under the transaction impact of different TPS under different orderer node arrangements. Figure. 8 shows the storage overhead changes of systems with different peer nodes and different orderer nodes under the transaction impact of stable 200tps.

Table. IV shows the effect of different system indexes on daily storage consumption when the system is stable(tps is in the range of 100 to 400). It can be concluded that for every thousand transaction(mainly certificates' status), one peer

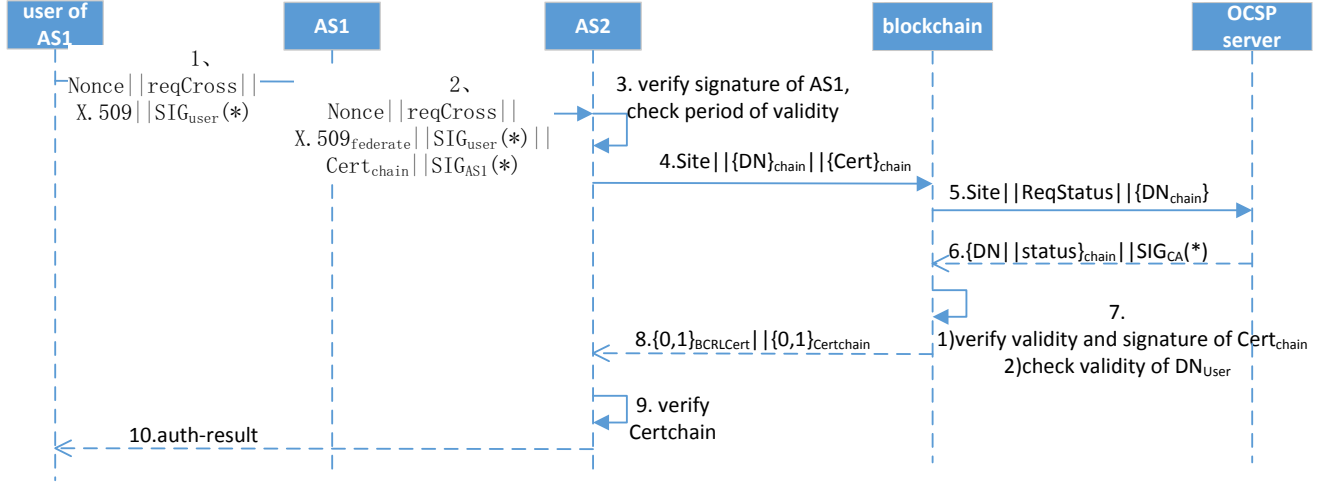


Figure 6. Cross-domain Authentication protocol

Table III
EXPERIMENTAL ENVIRONMENT

Operation System	Ubuntu 16.04
Network simulation tool	TC(Traffic Control), realize network environment such as network delay and packet loss
Hyperledger Fabric edition	1.1.0
blockchain node status	4 virtual machines,nodes (including orderer nodes and peer nodes) are variables, Kafaka consensus
Test smart contract	chain code-example02.go(official example)

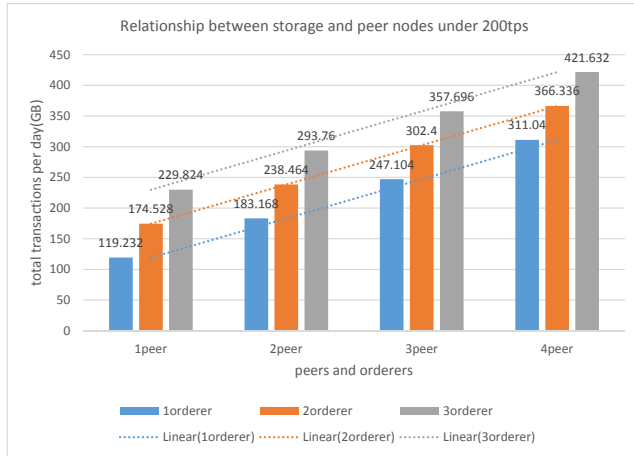


Figure 8. Relationship between storage and peer nodes under 200tps

node consumes about 3.2M and one orderer node consumes about 3.7M storage cost. The trust transfer system can basically meet the cross-domain authentication requirements of PKI systems and have better security.

Table IV
STORAGE COST UNDER STRESS TEST

daily data volume(GB)	Daily Transaction Volume (Thousands)	TPS (Transactions per second)	Data per thousand transactions(MB)	Number of orderer nodes	Number of peer nodes
155.52	8640	100	18	1	4
311.04	17280	200	18	1	4
466.56	25920	300	18	1	4
622.08	34560	400	18	1	4
183.168	8640	100	21.2	2	4
366.336	17280	200	21.2	2	4
549.504	25920	300	21.2	2	4
732.672	34560	400	21.2	2	4
210.816	8640	100	24.4	3	4
421.632	17280	200	24.4	3	4
632.448	25920	300	24.4	3	4
843.264	34560	400	24.4	3	4

VI. CONCLUSION

In this paper, we review the cross-domain authentication environment of PKI system and propose a blockchain-based scheme to realize trust enhancement of PKI systems. Our contributions are listed as follows: first, blockchain is introduced for PKI to verify the validity of the cert instead of CRL and OSCP. Then, we design the details of the cross domain protocol and further verify the systems efficiency. Finally, we proved the security of the proposed scheme in terms of the real-time issuance of certificate from the perspective of engineering. Our scheme will better meet the cross domain requirement when compared with traditional PKI solutions.

VII. ACKNOWLEDGEMENT

This research is supported by National Key Research and Development Program of China (2017YFB0802300)

and (2017YFB0802304). And by the major science and technology projects of Sichuan Province, research and development and application demonstration of key technologies for space security of Party and government information networks (2017GZDZX0002).

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